

Ayush Kumar

Full Stack Software Engineer

+91-7070472634 | Patna, Bihar | [Portfolio](#) | [Email](#) | [GitHub](#) | [LinkedIn](#)

Full Stack Engineer with **1 year of experience** in logistics and SaaS. Proficient in the **MERN stack**, **Next.js**, **TypeScript**, **React Native**, **Node.js**, **PostgreSQL**, and **RabbitMQ**. Built **RESTful APIs** and **microservice architectures**, achieving **25% faster API response times**, **40% better code reuse**, and **30% faster feature delivery**. Familiar with **Docker**, **AWS**, and **CI/CD pipelines**. An MBA background supports product thinking and Agile collaboration.

SKILLS

Languages: JavaScript (ES6+), TypeScript, SQL, C++, Python

Frontend: React.js, Next.js, React Native (Expo), HTML5, CSS3, Tailwind CSS, Redux Toolkit, Expo Router

Backend: Node.js, Express.js, RESTful APIs, FastAPI, WebSockets, Microservices Architecture, JWT, OAuth 2.0, RBAC

Databases: PostgreSQL, MongoDB, MySQL, Redis, Prisma ORM, Neon (Serverless PostgreSQL), Query Optimization, Indexing

DevOps & Cloud: Docker, Kubernetes, AWS (EC2, S3), CI/CD (GitHub Actions), Vercel, RabbitMQ, Kafka, Turborepo

Engineering: Data Structures & Algorithms, System Design, API Design, Caching, Performance Optimization, Testing

Tools: Git, GitHub, Postman, Linux, Agile/Scrum

Business: Product Thinking, Business Analysis, Stakeholder Communication, Cross-functional Collaboration

EXPERIENCE

Junior Software Engineer — NexoGrafix Private Limited

Apr 2026 – Present

- Developed a **Microsoft Word Add-in** in **TypeScript** using the **Office.js API**, automating document formatting and style enforcement against client-defined custom style guides — reducing manual effort by **60%**.
- Architected a **full-stack document conversion platform** (OCR + XML + EPUB pipeline) using **FastAPI** and **React**, decomposed into independent microservices for OCR processing, XML mapping, EPUB generation, and validation.
- Built the **XML** and **EPUB microservices** supporting JATS/DocBook schemas, configurable mapping templates, EPUB 3 generation with automated TOC, chapter splitting, CSS styling, and a validation engine with downloadable reports.

Software Engineer Intern — ShipU Logistics Private Limited

Sept 2025 – Jan 2026

- Designed a **logistics management platform** using **MERN** and **Next.js** with shipment tracking, delivery workflows, and **RBAC**; boosted backend performance by **25%** via optimised **PostgreSQL** queries and **Prisma ORM** connection pooling.
- Architected **event-driven microservices** using **RabbitMQ** and containerised services with **Docker**, deploying on **AWS** across development, staging, and production environments.
- Organised the codebase in a **Turborepo monorepo**, enabling shared modules across services and reducing duplicated code by **35%**.

Software Engineer Intern — Shabra Softech Solution Pvt. Ltd.

Feb 2025 – Jul 2025

- Migrated a legacy **MERN monolith** to a **Turborepo monorepo**, increasing code reusability by **40%** and reducing feature delivery time by **30%**.
- Built production-ready **Next.js** and **React Native** applications serving **1,000+ monthly active users**, alongside shared packages for UI components, Redux Toolkit state management, and a reusable API client.
- Implemented secure auth using **JWT**, **OAuth 2.0**, and **RBAC**, cutting authentication issues by **50%** while maintaining **95%+ on-time sprint delivery**.

PROJECTS

Aetheria — Real-Time 2D Multiplayer Sandbox World

[GitHub Link](#)

- Engineered a **real-time sync pipeline** using **Socket.io** and **Phaser Arcade Physics**, streaming updates at **20Hz** with client-side linear interpolation to guarantee smooth character rendering at **60 FPS**.
- Built interactive sandbox features including modular wardrobe customisers, transactional item shops, **WebRTC** proximity signalling for peer-to-peer voice feeds, and private room decoration grids with drag-and-drop rotation and coordinate persistence.
- Integrated real-time synced multiplayer mini-games (Chess, Tic-Tac-Toe, Connect Four) and single-player retro games (Snake, Memory Match) directly in-canvas, rewarding players with experience points and in-game economy currencies.
- Designed a zero-friction database layer using **Prisma ORM** connection pools with serverless **Neon PostgreSQL** adapters, implementing an automated zero-config fallback to in-memory arrays for fast local development.

- Created admin auditing utilities (AdminPanel.tsx) monitoring live sockets, server uptime, active player sessions, database records, and role hierarchy overrides.

Excalidraw Clone — Real-Time Collaborative Whiteboard

GitHub Link

- Built a **real-time collaborative whiteboard** using **Next.js**, **WebSockets**, and **Turborepo**, supporting multiple users per room with low-latency synchronisation.
- Applied persistent storage using **PostgreSQL** with optimised queries, reducing canvas loading time by **40%**.
- Improved rendering performance by applying **React memoisation and state diffing**, reducing unnecessary re-renders by **60%** during concurrent editing.

RideSync — Real-Time Ride Booking Application

GitHub Link

- Built a **full-stack ride-hailing application** using **React Native (Expo)** with real-time GPS tracking and WebSocket-based communication.
- Integrated **Clerk authentication** and **Neon serverless PostgreSQL**, achieving average database response times under **200 ms**.
- Implemented global state management with **Zustand** and routing with **Expo Router**; optimised bundle size by **15%** through lazy loading and code splitting.

EDUCATION

2026 – Present	Master of Computer Applications (MCA) IIIT Ranchi & IIT Patna	
2023 – 2025	Master of Business Administration (MBA) — Marketing Impact College, Aryabhata Knowledge University	CGPA: 8.61 / 10
2020 – 2023	Bachelor of Science (Honours) — Chemistry B.D. College, Patliputra University	60.4%